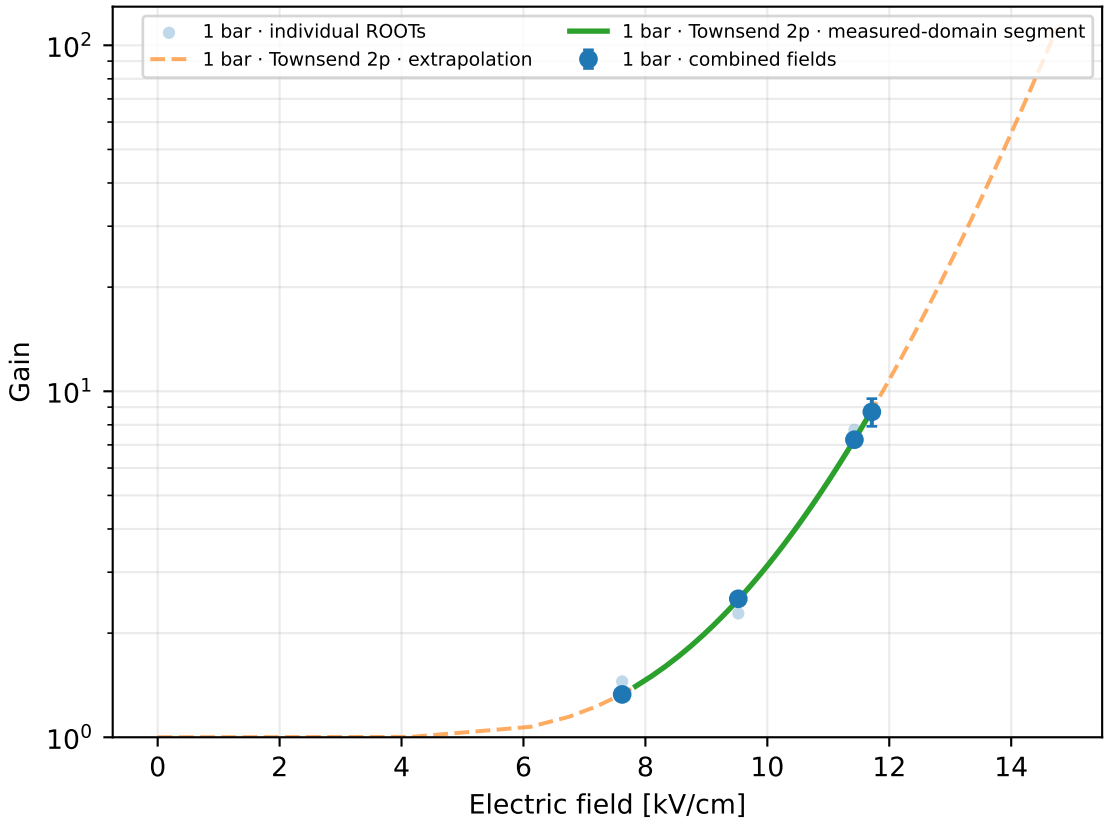
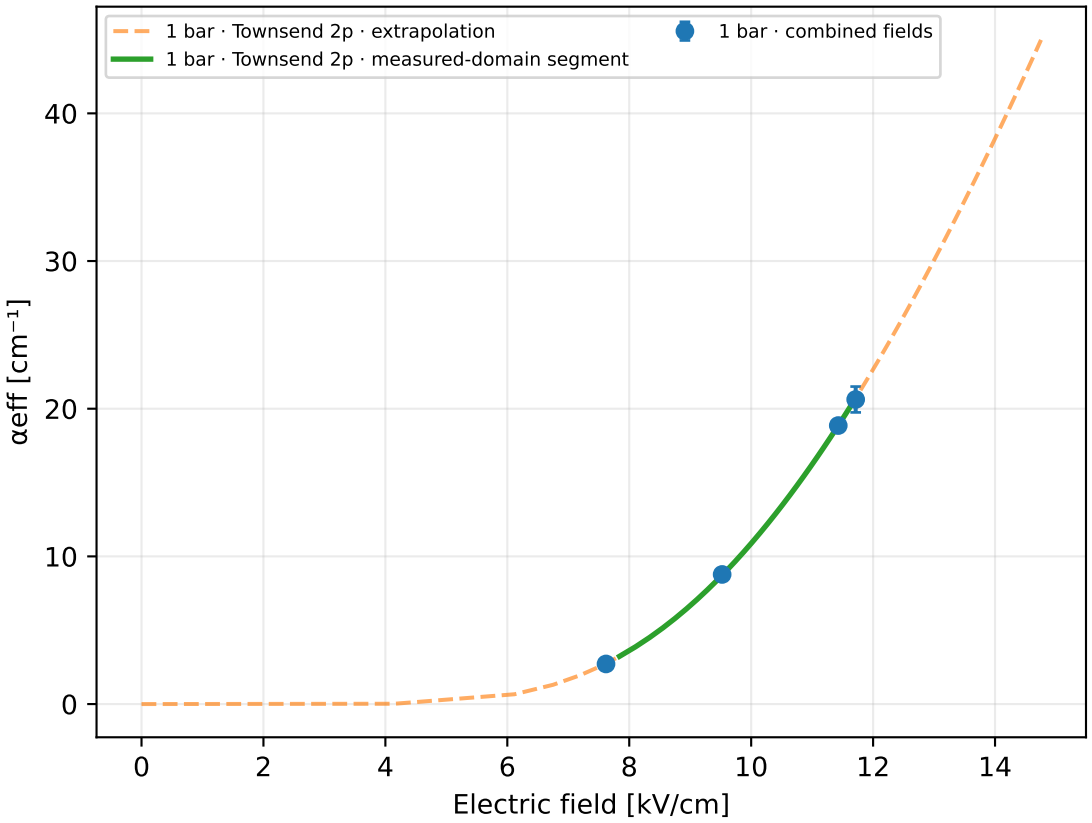
ArN2 · Ar 99.5 % / N₂ 0.5 % · gap 1.05 mm

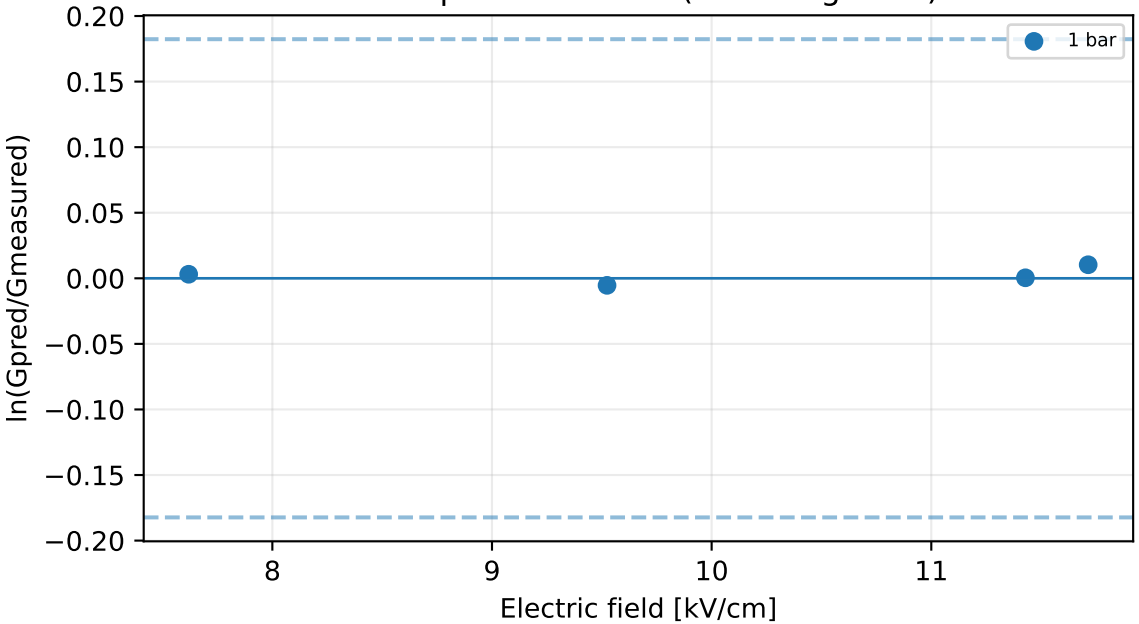
Gain vs electric field



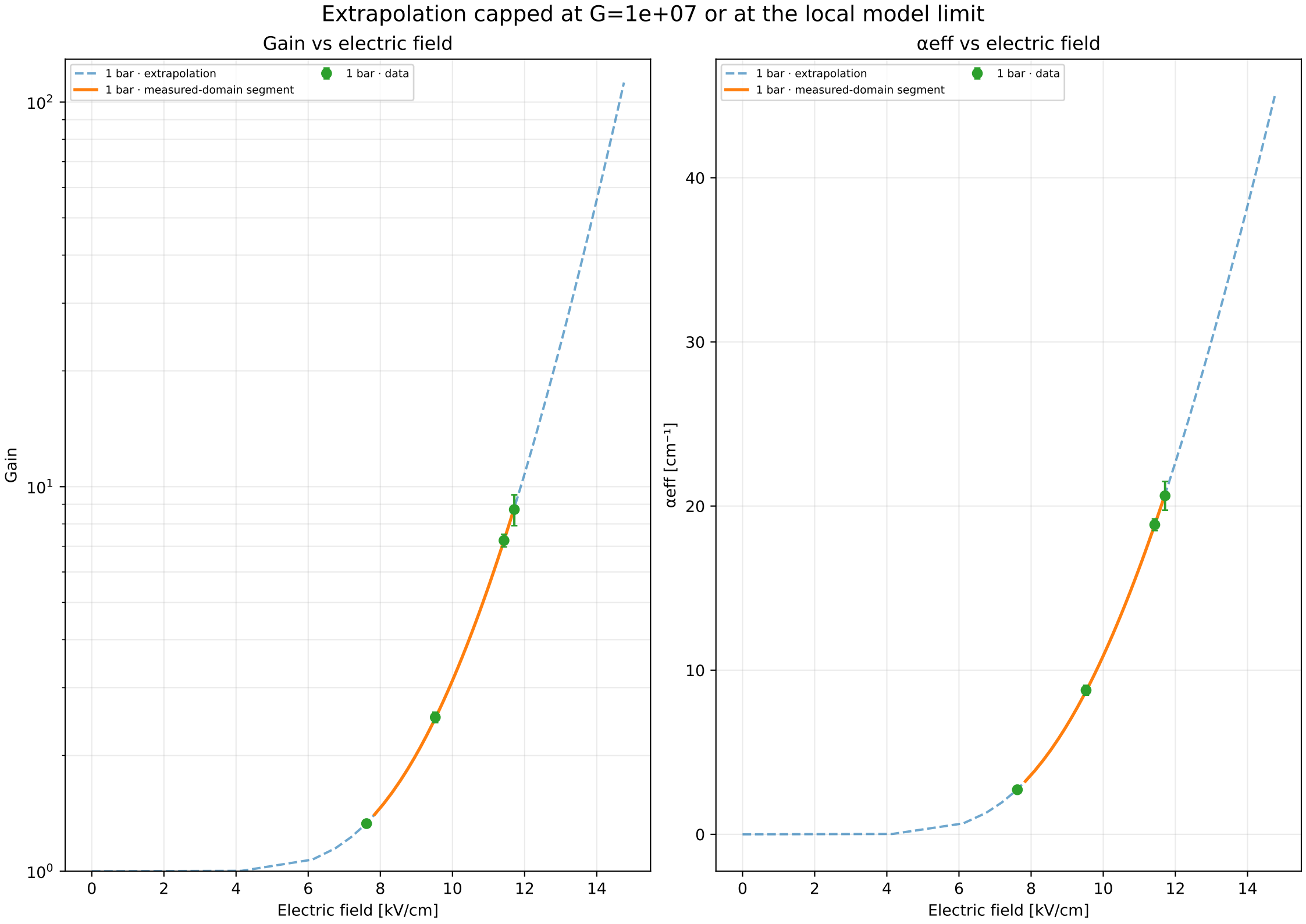
Effective Townsend coefficient vs electric field



Gain-space residuals (±ln 1.2 guides)




```
ROOT runs: 7
Unique physical points: 4
Pressures: 1 bar
Selected model: Townsend 2p
Status: rejected_poor_cross_validation
Training RMSE: ×1.007
Cross-validation median: unavailable
Cross-validation worst: unavailable
Reduced-field range: 7.619–11.71 kV cm⁻¹ bar⁻¹
A=890.59, B=44.053, m=0, n=1
Covariance condition: 1.12e+03
Warnings: poor_cross_validation, large_cross_validation_outlier
1 bar display limit: G=112 (local inversion limit)
```


Extrapolation capped at G=1e+07 or at the local model limit
Gain vs electric field
αeff vs electric field
1 bar · extrapolation
1 bar · measured-domain segment
1 bar · data
Gain
αeff [cm⁻¹]
Electric field [kV/cm]
Electric field [kV/cm]

Exact combined points used by the fit · page 1/1

Repeated ROOTs at the same field are pooled once; all source ROOTs remain on disk.

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm⁻¹]	npe	runs
1	7.61905	7.61905	1.33	0.0269	2.71599	600	2
1	9.52381	9.52381	2.513333	0.0789	8.77724	600	2
1	11.4286	11.4286	7.245	0.274	18.8601	600	2
1	11.7143	11.7143	8.72	0.801	20.6249	100	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-24.2	—	—	no	poor cross validation, large cross validation outlier